

CAPSTONE RESEARCH PROJECT

Forecasting Corporate Credit Spreads Using Macroeconomic & Financial Indicators

An Expanding-Window Regression Framework for One-Month-Ahead Forecasting

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Springboard Data Science Track

M.S. in Business Analytics

- Target: BAA Yield – 10Y Treasury
- Data: FRED / ALFRED Revisions
- Horizon: 1-Month Ahead

Problem Statement & Research Question

Corporate Credit Spreads

Corporate credit spreads reflect **the additional yield** investors demand for holding corporate debt relative to safer government securities. They typically widen during periods of economic and financial stress and tighten as conditions improve.

The Challenge: High Persistence

Credit spreads **are highly persistent**, so a simple forecast based on the current spread already produces relatively low forecast error. Finding predictive indicators that improve upon this baseline is a major forecasting hurdle.

RESEARCH QUESTION:

Can one-month-ahead credit spread forecasts be improved by incorporating macroeconomic releases, financial market indicators, oil prices, and the spread's own historical dynamics?

Data & Target Variable

Data Source

Federal Reserve Economic Data (FRED) and Archival FRED (ALFRED),
Federal Reserve Bank of St. Louis

Yahoo Finance—market data

Target Variable

**Moody's BAA Corporate Bond Yield
– 10-Year U.S. Treasury Yield**

Represents the additional yield investors demand for holding BAA-rated corporate debt relative to U.S. Treasuries

FREQUENCY

Monthly

FORECAST HORIZON

1 Month Ahead

SAMPLE PERIOD

Feb 1991 - Jul 2026

Predictors

Macroeconomic

- **Inflation (YoY)**
Price pressures & economic conditions
- **Unemployment Rate**
Labor-market strength
- **GDP Growth (YoY)**
Broad economic growth
- **Industrial Production (YoY)**
Growth in manufacturing & industrial output
- **Federal Funds Rate**
Monetary-policy & borrowing costs

Interest Rates

- **10-Year Treasury Yield**
Long-term interest-rate conditions
- **10Y-3M Yield Curve Slope**
Economic outlook & recession expectations

Market Conditions

- **VIX**
Market uncertainty & risk aversion
- **S&P 500 Returns**
Broad U.S. equity performance
- **Russell 2000 Returns**
Small-cap credit sensitivity
- **WTI Oil Returns**
Energy-cost & inflation pressures

Spread Dynamics

- **Credit Spread Volatility**
Recent variability and instability in credit spreads
- **Lagged Credit Spread**
Persistence in historic credit spread levels

Data Preparation and Temporal Alignment

Data Collection

FRED/ALFRED

Daily, monthly, quarterly series

Release-Aware Processing

Initial-release macroeconomic data used where available.

Release timing preserved

Frequency Alignment

Series standardized to monthly frequency.

Daily and quarterly data aligned appropriately

Temporal Alignment

Predictors aligned to information available at each forecast date. Observations assigned to **month-end timestamps**.

Key Principle

Release-aware alignment minimizes look-ahead bias by using only information available at each point in time.

Feature Engineering

Macroeconomic

- **CPI → Inflation (YoY)**
12-month percentage change
- **Industrial Production → Growth (YoY)**
12-month percentage change
- **Real GDP → Growth (YoY)**
Year-over-year percentage change

Market Transformations

- **S&P 500 → Returns**
Based on month-end closing
- **Russell 2000 → Returns**
Based on month-end closing
- **WTI Crude Oil → Returns**
Based on month-end values
- **VIX → Monthly Average**
Average of daily observations

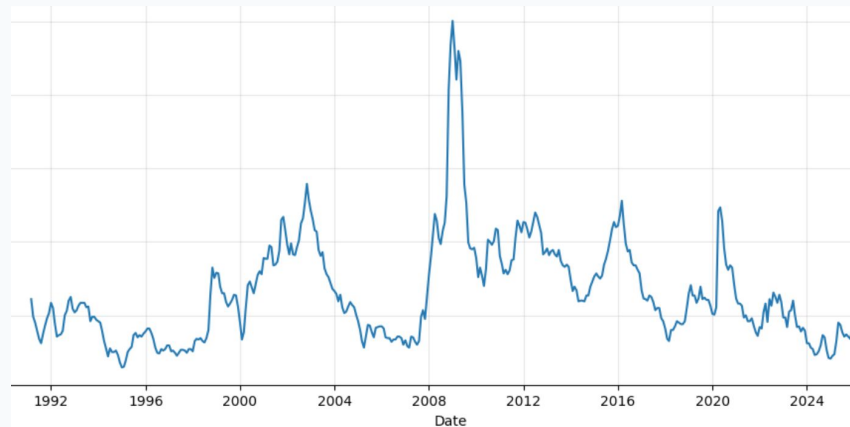
Spread Dynamics

- **Credit Spread → 6-Month Rolling Volatility**
Rolling standard deviation of credit spread

Exploratory Data Analysis

Credit Spread Behavior Over Time

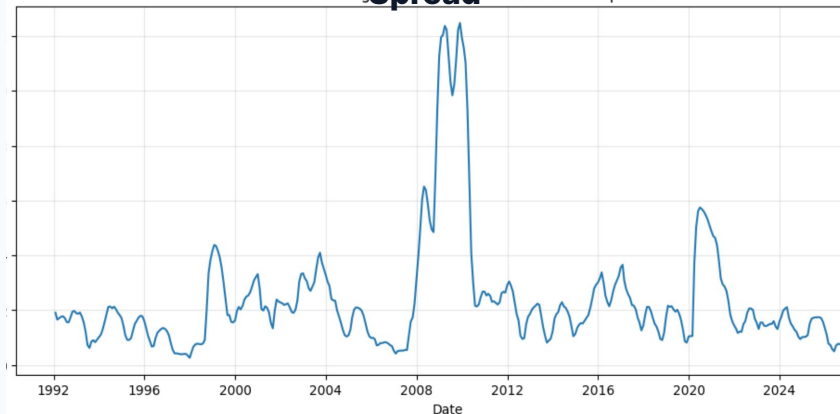
BAA-Treasury Credit Spread Over Time



Credit Spread Level

Spreads widen sharply during periods of financial and economic stress, with the largest increase occurring during the 2008–09 Global Financial Crisis.

12-Month Rolling Standard Deviation of Credit Spread



Credit Spread Volatility

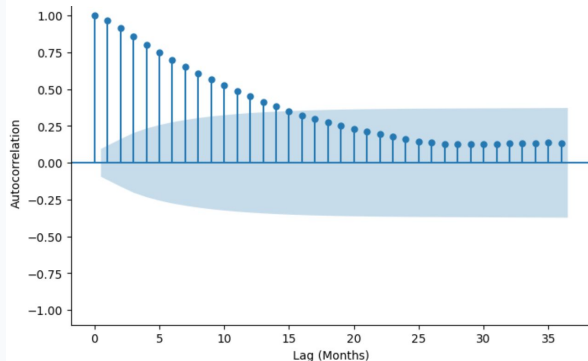
Rolling volatility rises substantially during periods of market stress, particularly during the 2008–09 financial crisis.

This motivated the inclusion of recent credit-spread volatility as a forecasting feature.

Exploratory Data Analysis

Time-Series Diagnostics

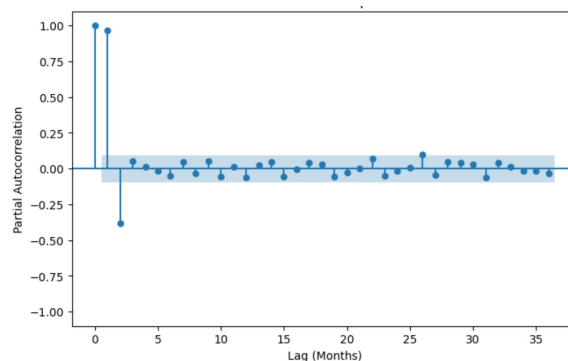
ACF of Credit Spread



Autocorrelation Function

- Strong autocorrelation at short lags indicates substantial persistence in credit spreads.
- Recent spread levels contain information about future spread levels.

PACF of Credit Spread



Partial Autocorrelation

- Strong direct relationships are concentrated in the first two lags.
- Lag 1 is strongly positive, while lag 2 shows a notable negative relationship after controlling for lag 1.

ADF Stationarity Test

PURPOSE

Test for unit root

HYPOTHESIS (H_0):

The credit spread series has a unit root

RESULT

Reject H_0 ($p = 0.0032$)

CONCLUSION

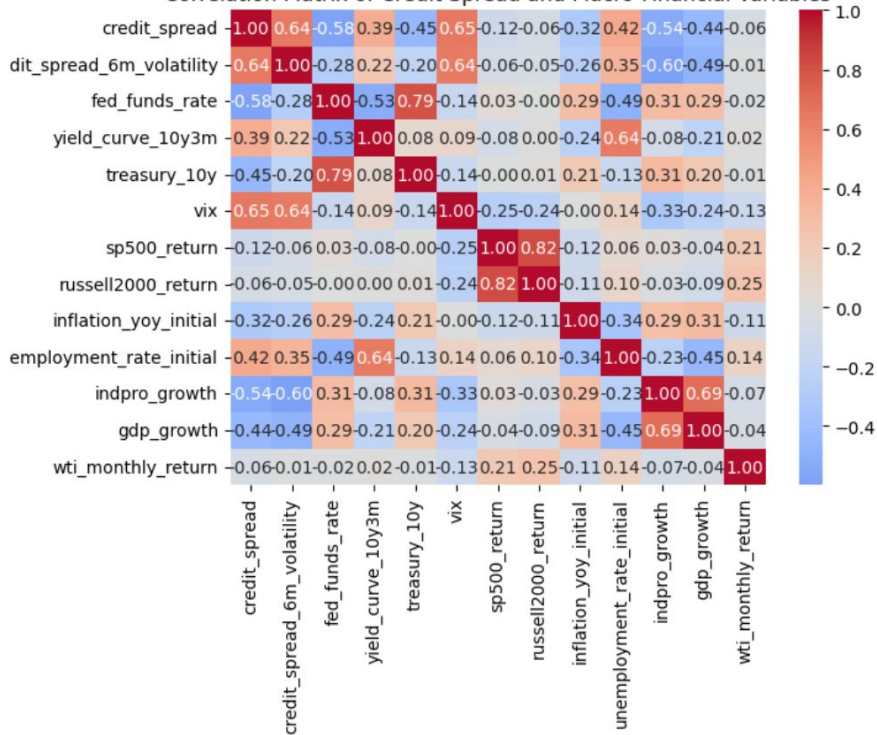
Evidence supports stationarity

Supports modeling the credit spread in levels.

Exploratory Data Analysis

Correlation Analysis & Lagged Relationships

Correlation Matrix of Credit Spread and Macro-Financial Variables



Correlation Analysis

Strongest positive:
VIX (+0.65) & spread volatility (+0.64)

Economic:
Unemployment (+0.42), industrial prod. (-0.54), GDP (-0.44)

Interest rates:
Fed funds (-0.58), 10Y Yield (-0.45)

Equity returns:
S&P 500 (-0.12), Russell (-0.06) (weak)

	Lag 0	Lag 1	Lag 3	Lag 6	Lag 12
fed_funds_rate	-0.58	-0.55	-0.50	-0.42	-0.24
yield_curve_10y3m	0.39	0.37	0.35	0.33	0.18
treasury_10y	-0.45	-0.44	-0.39	-0.32	-0.24
vix	0.65	0.66	0.57	0.45	0.29
sp500_return	-0.12	-0.22	-0.25	-0.22	-0.14
russell2000_return	-0.06	-0.16	-0.18	-0.15	-0.10
wti_monthly_return	-0.06	-0.11	-0.12	-0.05	-0.00
indpro_growth	-0.54	-0.50	-0.41	-0.30	-0.11
gdp_growth	-0.44	-0.40	-0.34	-0.24	-0.16
inflation_yoy_initial	-0.32	-0.27	-0.20	-0.13	-0.16
unemployment_rate_initial	0.42	0.38	0.32	0.24	0.14
credit_spread_6m_volatility	0.64	0.61	0.54	0.39	0.28

Lagged Relationships

- **Most relationships weaken as the lag increases**
- **Equity returns peak around Lag 3:** S&P 500 (-0.25), Russell 2000 (-0.18)
- **WTI strengthens through Lag 3** (-0.12), but remains weak

Forecast Feature Construction

Building the One-Month-Ahead Modeling Dataset

Forecast Target

Information at **Month t** → Credit Spread at **Month t+1**

`credit_spread_target = credit_spread.shift(-1)`

Shifts the credit spread one month backward so each month's predictors are paired with the following month's spread.

Predictor Lags

Lag 1 | **Lag 3** | **Lag 6** | **Lag 12**

Lagged versions of predictors were created to capture delayed relationships.

Feature-Target Correlations

	credit_spread_target
credit_spread	0.97
credit_spread_lag1	0.92
credit_spread_lag3	0.81
vix	0.66
credit_spread_lag6	0.66
vix_lag1	0.62
credit_spread_6m_volatility	0.61
credit_spread_6m_volatility_lag1	0.58
fed_funds_rate	-0.55
vix_lag3	0.53
fed_funds_rate_lag1	-0.53
indpro_growth	-0.50
credit_spread_6m_volatility_lag3	0.49
fed_funds_rate_lag3	-0.47
indpro_growth_lag1	-0.45

Key Findings:

Strong Target

Persistence: Current credit spread has the strongest correlation with next month's spread (0.97), followed by Lag 1 (0.92) and Lag 3 (0.81)

Additional Relationships

Macroeconomic, market, and interest-rate variables and their lags also show associations with the next-month target.

Forecast Evaluation Framework

Time-Based Split, Forecasting Benchmarks, Expanding Window Approach

Time-Based Split

Training / Development

Feb 1991 – Dec 2021

Out-of-Sample Test

Jan 2022 – Jul 2026

Chronological split preserves time order and prevents future information from entering model training.

Forecasting Benchmarks

Naïve Persistence

Next month's spread = current spread

MAE 0.0931 | RMSE 0.1184

AR(1) Autoregressive Model

Fits next month's spread from current spread via fitted intercept & persistence

MAE 0.0948 | RMSE 0.1185

$\text{Spread}_{t+1} = 0.07 + 0.9696 \times \text{Spread}_t$

AR(1) coefficient ≈ 0.97 indicates very strong month-to-month persistence in credit spreads.

Expanding Window

Develop regression models using an expanding-window framework to test against benchmarks.

Evaluation Loop

1. Train Model
2. Forecast next month
3. Add observation
4. Re-estimate & Repeat

Uses all information available up to each forecast date.

Regression Model Development & Performance

Progressive OLS Specifications & Out-of-Sample Results

1. Baseline OLS – Economic/Market Predictors

External economic & market predictors only (no credit spread included).

Underperformed benchmarks; excluded from final comparison.



2. OLS – Spread + Economic / Market Predictors

Current credit spread + current economic & market predictors



3. Final OLS – Current +Lagged

Selected current & lagged credit spread + macro-financial predictors

Out-of-Sample Model Performance Comparison

	Model	MAE	RMSE	R ²
0	Naive Persistence	0.0931	0.1184	0.7666
1	AR(1)	0.0948	0.1185	0.7664
2	OLS – Spread + Economic/Market Predictors	0.0854	0.1057	0.8140
3	Final Expanding-Window OLS	0.0768	0.0993	0.8359

Final OLS Feature Set

Spread Dynamics: Credit Spread (current, L1) · 6M Spread Volatility

Market Risk: VIX

Interest Rates: Fed Funds · 10Y Treasury (current, L1) · 10Y–3M Yield Curve

Equity Markets: S&P 500 Return (current, L3) · Russell 2000 Return (L1, L3)

Macroeconomic: Inflation (YoY) · Unemployment · Industrial Production (YoY) · GDP Growth (YoY)

Commodity: WTI Oil Return (L3)

Directional Forecast Evaluation

Can the model anticipate spread widening vs tightening?

Direction Definition: $Direction_t = \text{sign}(\text{Spread}_{t+1} - \text{Spread}_t)$

+1 = Widening | -1 = Tightening | 0 = Unchanged

Directional Accuracy (DA) = (Correct Direction Predictions) / Total Forecasts

Model	DA	Predicted Tightening	Predicted Widening	Predicted Unchanged
AR(1)	43.64%	0	55	0
Final OLS	65.45%	29	26	0
Actual	—	29 Tightening	24 Widening	2 Unchanged

Pesaran-Timmermann Test

Statistical Predictability Test

PT Statistic: **2.59**

p-value: **0.0097**

Significant at 1% level

Provides evidence that the model's directional forecasts contain predictive information beyond chance.

Statistical Comparison of Forecast Accuracy

Diebold–Mariano Test: Final OLS vs. Naïve Persistence

Diebold–Mariano (DM) Test

Tests whether the difference in forecast accuracy between two models is statistically significant rather than due to sampling variation.

Is the Final OLS consistently more accurate than naïve persistence?

H_0 : Equal expected forecast accuracy

H_1 : Different expected forecast accuracy

Squared forecast errors are compared across the 55 out-of-sample forecasts.

Results & Interpretation

Average Loss Difference: -0.004163

DM Statistic = -1.8142

p-value = 0.0697

Significant at 10%, but not 5%

Negative loss difference → Final OLS had lower squared forecast error on average.

The results provide **moderate evidence** that the Final OLS improves point-forecast accuracy relative to naïve persistence, although the improvement is not significant at the conventional 5% level

Conclusions & Future Work

Conclusions

- Combining current & lagged macro-financial information with credit-spread dynamics improved one-month-ahead forecasts beyond benchmarks.
- 17.5% lower MAE & 16.1% lower RMSE vs. naive persistence ($R^2 = 0.836$)
- 65.45% directional accuracy vs. 43.64% for AR(1)
- Moderate statistical evidence for point-forecast accuracy, but not significant at 5% level (DM $p = 0.0697$)

Limitations

- **Data Availability & Sample Size:** Point-in-time availability was constrained; final evaluation had 55 monthly forecasts.
- **Temporal Alignment:** Avoiding look-ahead bias may have lagged some predictors more than necessary.
- **Specification Search:** Not all potential predictors and lag combinations were systematically evaluated.

Future Work

- **Model Validation:** Use separate training, validation, and final holdout periods for feature selection.
- **Broader Feature & Optimization:** Systematically evaluate additional predictors and lag structures.
- **Stress-Period Evaluation:** Evaluate forecast performance separately during periods of financial and economic stress.
- **Expand Point-in-Time Dataset:** Incorporate additional release-aware macroeconomic and financial indicators.